

Mock Paper 1C — Year 12 Mock (Week 27, UNSEEN)

Time allowed: 1 hour 30 minutes · Total: 70 marks · Answer all questions.

TEACHER ONLY until sat. This paper is unseen: do not release it, set it as homework, or file it in the student-facing library before Mock Week (Week 27). It covers exactly what Year 12 has taught by then: 1.1–1.4 complete, all of 2.1, and 2.2.1.

Instructions: Show all working on calculations and logic questions. Aim for about 1 minute per mark and leave time to check.

AO key: **AO1** = knowledge & understanding · **AO2** = application · **AO3** = analysis/evaluation/design.

Questions

Question 1 — Processors (7 marks)

1(a) State the role of the **program counter (PC)** and the **memory address register (MAR)** during the fetch stage of the fetch–decode–execute cycle. **[2]**

1(b) Explain **one** way that increasing the amount of **cache** can improve the performance of a processor. **[2]**

1(c) A drone's flight controller uses a battery-powered processor with a **RISC** instruction set and a **Harvard** architecture. Explain why this combination is suitable for the drone. **[3]**

(Running total: 7)

Question 2 — Storage (4 marks)

2(a) A hospital must keep copies of old patient scans for 30 years. The scans are very large and are almost never accessed. State the **most appropriate** storage technology and justify your choice. [2]

2(b) State what is meant by **virtual storage** and give **one** drawback of relying on it. [2]

(Running total: 11)

Question 3 — Systems software and translators (6 marks)

3(a) An operating system uses **round robin** scheduling. Describe how round robin scheduling works and state **one** advantage it has over first come first served. [3] (AO1/AO2)

3(b) State the **four stages of compilation** in the order they are carried out. [2]

3(c) State **one** reason a developer might choose to write part of a program in **assembly language** rather than a high-level language. [1]

(Running total: 17)

Question 4 — Software development methodologies (6 marks)

A two-person independent studio is building a mobile puzzle game. The requirements will keep changing in response to player feedback, the budget is small, and the studio wants a playable version released early and updated often.

4 Recommend a software development methodology for the studio. Justify your recommendation with reference to the scenario, and evaluate it against **at least one** alternative methodology. [6]

(Running total: 23)

Question 5 — Databases (6 marks)

A dental surgery stores data in the following tables:

```
Patient(PatientID, Surname, DateOfBirth)
Appointment(AppointmentID, PatientID, ApptDate, RoomNumber)
```

5(a) Identify the **primary key** and the **foreign key** of the Appointment table. [2]

5(b) Write an SQL statement to display the Surname and DateOfBirth of every patient born after 1 January 2005, sorted into alphabetical order of surname. [3]

5(c) State what is meant by **referential integrity**. [1]

(Running total: 29)

Question 6 — Networks and web technologies (6 marks)

6(a) When data is sent across the internet it is split into **packets**. State **two** items of information carried in a packet's header. [2]

6(b) Describe the role of the **DNS** when a user types a URL into a browser. [2]

6(c) A booking form is validated by **JavaScript in the browser** before it is submitted. Explain why the server must **repeat** this validation even though the client has already performed it. [2]

(Running total: 35)

Question 7 (starred) — Compression, encryption and hashing (9 marks)

The quality of your extended response will be assessed in this question.

PhotoVault is a new cloud photo-sharing service. Users upload photos from their phones over the internet, view them in a web gallery, and can pay for a "vault" that guarantees their original photos are preserved exactly. Users log in with a username and password.

7 Discuss how PhotoVault should use **compression**, **encryption** and **hashing** in its service. In your answer you should consider the choice between lossy and lossless compression, the protection of data both in transit and at rest, and the safe storage of passwords. Reach justified recommendations. [9]

(Running total: 44)

Question 8 — Data types and representation (6 marks)

8(a) Convert **−90** to 8-bit two's complement binary. Show your working. [2]

8(b) Convert the denary number **172** to hexadecimal. [1]

8(c) A floating point number is stored with a 6-bit two's complement **mantissa** of **010110** and a 4-bit two's complement **exponent** of **0011**. Calculate its denary value. Show your working. [2]

8(d) State **one** advantage of **Unicode** over ASCII. [1]

(Running total: 50)

Question 9 — Data structures and Boolean algebra (7 marks)

9(a) The following operations are carried out on an empty **stack**: `push(4)`, `push(7)`, `pop()`, `push(2)`, `push(5)`, `pop()`. State the value returned by **each** `pop()`, and the contents of the stack at the end, identifying the top item. [2]

9(b) State why a **queue** is an appropriate data structure for managing jobs sent to a shared printer. [1]

9(c) Complete the truth table for the expression $Q = (A \text{ OR } B) \text{ AND } \text{NOT}(A \text{ AND } B)$. [2]

A	B	A OR B	A AND B	Q
0	0			
0	1			
1	0			
1	1			

9(d) Simplify the Boolean expression $A \text{ OR } (A \text{ AND } B)$, naming the law you have used. [2]

(Running total: 57)

Question 10 — Computational thinking (6 marks)

A supermarket's self-checkout terminals currently look up every scanned product's price from a central database, and each step of a sale (scanning, card payment, receipt printing) runs one after another. At busy times the terminals are slow. The developers propose (i) **caching** the day's price list on each terminal and (ii) running the steps of a sale **concurrently** where possible.

10 Discuss the benefits and drawbacks of the two proposals, and reach a justified conclusion about whether the developers should adopt them. [6]

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(Running total: 63)

Question 11 — Programming techniques (7 marks)

11(a) Write a function `countVowels(word)` in **pseudocode** that takes a string parameter `word` and returns the number of vowels (a, e, i, o, u) it contains. Your function must work for both upper-case and lower-case letters. [4]

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11(b) State the difference between a **function** and a **procedure**. [1]

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11(c) Explain the difference between passing a parameter **by value** and **by reference**. [2]

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(Running total: 70)
